# Exercises for Chapter 3, first part, with solutions

**Graph Algorithms**

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The exercises cover some of the material covered in Chapter 3. Each exercise is of one of the following types: basic, suggested, or more challenging. Basic exercises deal with concepts from lectures and are intended to check understanding of basic concepts. Suggested exercises deal with concepts from lectures and are similar to exercises that you may expect on exams. More challenging exercises address concepts from lectures, but these exercises are less likely to appear on exam or are intended for a higher grade.

1. *(Decision versions of optimization problems; basic.)*

   Formulate the decision versions of the following optimization problems: MINIMUM SPANNING TREE, MAXIMUM FLOW, MAXIMUM MATCHING and MINIMUM VERTEX COVER IN BIPARTITE GRAPHS.

   **Solution:** For simplicity, let us agree that, when converting an optimization problem into a decision problem, for the additional input data we take *an integer* if the value of the optimal solution of the optimization problem is always an integer, otherwise we take *a real number.*

   According to this agreement, we transform the given four optimization problems into decision problems as follows:

| MINIMUM SPANNING TREE | |
|---|---|
| **Input:** | A connected graph $G = (V, E)$, a weight function $w : E \to \mathbb{R}$, a real number $r$. |
| **Question:** | Is there a spanning tree $T$ in the graph $G$ such that $w(T) \leq r$? |

| MAXIMUM FLOW | |
|---|---|
| **Input:** | A network $N = (D, c, s, t)$, a real number $r$. |
| **Question:** | Is there a flow $f$ in the network $N$ such that $val(f) \geq r$? |

| MAXIMUM MATCHING | |
|---|---|
| **Input:** | A graph $G = (V, E)$, an integer $k$. |
| **Question:** | Does the graph $G$ have a matching $M$ such that $\lvert M\rvert \geq k$? |

| MINIMUM VERTEX COVER IN BIPARTITE GRAPHS | |
|---|---|
| **Input:** | A bipartite graph $G = (V, E)$, an integer $k$. |
| **Question:** | Does the graph $G$ have a vertex cover $C$ such that $\lvert C\rvert \leq k$? |

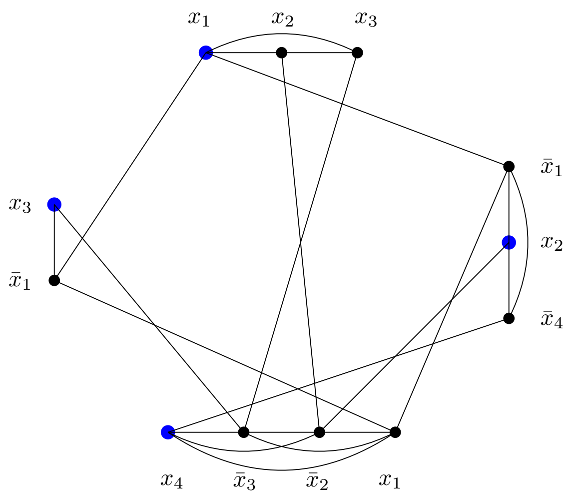
2. *(NP-completeness of* Independent Set*; basic)*

Draw the graph of $G$ obtained by using the polynomial reduction (from the lectures) of the Satisfiability problem to the Independent Set problem if the input data for Satisfiability is given by the following formula:

$$\varphi = (x_1 \vee x_2 \vee x_3) \wedge (\bar{x}_1 \vee x_2 \vee \bar{x}_4) \wedge (x_1 \vee \bar{x}_2 \vee \bar{x}_3 \vee x_4) \wedge (\bar{x}_1 \vee x_3)\,.$$

Does the graph $G$ contain 4 pairwise nonadjacent vertices? What can you conclude from this about the formula $\varphi$?

**Solution:** The graph $G$ is shown in the following figure.




Yes, the graph $G$ contains 4 pairwise non-adjacent vertices (for example, the four vertices marked in blue in the figure). From this, we conclude that the statement $\varphi$ is satisfiable.

Each independent set of cardinality 4 corresponds to a set of values $\top, \bot$ for the variables $x_1, x_2, x_3, x_4$ for which all clauses of the formula $\varphi$ take value $\top$. For example, the independent set in the figure corresponds to setting $x_i = \top$ for all $i \in \{1, 2, 3, 4\}$.

3. *(Proving NP-completeness in general; suggested.)*

Let $\Pi$ and $\Pi_1$ be decision problems such that problem $\Pi$ belongs to the class NP, problem $\Pi_1$ is NP-complete, and $\Pi_1 \propto \Pi$. Prove that the problem $\Pi$ is NP-complete.

**Solution:** To prove that problem $\Pi$ is NP-complete, we verify that it belongs to NP and is NP-hard. That the problem $\Pi$ belongs to the class NP is already true by assumption.

Let us show that the problem $\Pi$ NP-hard. By definition, that means yes the existence of a polynomial-time algorithm for the $\Pi$ problem implies $\mathsf{P} = \mathsf{NP}$. So let us assume that there is a polynomial-time algorithm $A$ for the problem $\Pi$. Since $A$ is a polynomial-time algorithm, there exists a positive integer $k$ such that algorithm $A$ runs in $\mathcal{O}(n^k)$ time on input data of size at most $n$. We want to show that $\mathsf{P} = \mathsf{NP}$.

By assumption, $\Pi_1 \propto \Pi$ holds, i.e., there is a polynomial reduction of the problem $\Pi_1$ to the problem $\Pi$. In other words, there exists a polynomial-time algorithm $A_1$ that, for any instance $I$ of $\Pi_1$, constructs an instance $J = J(I)$ for the problem $\Pi$ such that the answer to $\Pi_1$ at $I$ is the same as the answer to $\Pi$ at $J$. Since $A_1$ is a polynomial-time algorithm, there exists a positive integer $\ell$ such that algorithm $A_1$ runs in $\mathcal{O}(n^\ell)$ time on input data of size at most $n$.

Let us show that there is a polynomial-time algorithm for the problem $\Pi_1$. Given an instance $I$ of the problem $\Pi_1$, we first use algorithm $A_1$ to compute in time $\mathcal{O}(|I|^\ell)$ an instance $J = J(I)$ for the problem $\Pi$ such that the answer to $\Pi_1$ at $I$ is the same as the answer to $\Pi$ at $J$. The problem $\Pi$ can be solved on the input $J$ in polynomial time with the algorithm $A$. Since the answer to $\Pi_1$ at $I$ is the same as the answer to $\Pi$ at $J$, we have also solved the $\Pi_1$ problem. The algorithm described in this way is a polynomial-time algorithm for the $\Pi_1$ problem, since $|J| = \mathcal{O}(|I|^\ell)$ and the execution time of the described algorithm can be estimated as $\mathcal{O}(|J|^k) = \mathcal{O}((|I|^\ell)^k) = \mathcal{O}(|I|^{k\ell})$. Thus, we have shown that there is a polynomial-time algorithm for the problem $\Pi_1$.

By assumption, the problem $\Pi_1$ $\mathsf{NP}$-complete. This means that the existence of a polynomial-time algorithm for problem $\Pi_1$ implies $\mathsf{P} = \mathsf{NP}$. We have shown that there is a polynomial-time algorithm for the $\Pi_1$ problem. It follows that $\mathsf{P} = \mathsf{NP}$, which we wanted to show.

4. *(Proving* NP*-completeness; suggested.)*

   Show that the following problem is NP-complete.

| | |
|---|---|
| CODOMINATING SET | |
| **Input:** | A graph $G = (V, E)$, an integer $k$. |
| **Question:** | Is there a set $D \subseteq V$ in the graph $G$ such that $\|D\| \leq k$ and there does not exist a vertex $v \in V \setminus D$ such that $D \subseteq N(v)$? |

*Hint: Describe the given property of a set $D$ in the complement of the graph $G$.*

**Solution:** Let us first name the property of a set that the problem CODOMINATING SET asks about. A set $D \subseteq V$ is a *codominating set* in the graph $G = (V, E)$, if there is no vertex $v \in V \setminus D$ such that $D \subseteq N(v)$. In the lectures, we derived the NP-completeness of CLIQUE by modeling with it the INDEPENDENT SET problem using the graph $\overline{G}$ (that is, in the complement of the graph $G$).[1] Let us follow the hint and use a similar approach here. Let us show that $D$ is a codominating set in the graph $G$ if and only if $D$ is a dominating set in the graph $\overline{G}$.[2]

Let us write a sequence of pairwise equivalent conditions for the set $D$:

(i) $D$ is a codominant set in the graph $G$.

(ii) For every vertex $v \in V \setminus D$, $D \nsubseteq N_G(v)$ holds.

(iii) For every vertex $v \in V \setminus D$ there exists a vertex $w \in D$ that is not adjacent to the vertex $v$ in the graph $G$.

(iv) For every vertex $v \in V \setminus D$ there exists a vertex $w \in D$ that is adjacent to the vertex $v$ in the graph $\overline{G}$.

(v) $D$ is a dominating set in the graph $\overline{G}$.

[1] Recall that $V(\overline{G}) = V$ and two distinct vertices $x, y \in V$ are adjacent in the graph $\overline{G}$ if and only if they are not adjacent in the graph $G$.

[2] Recall that a *dominating set* in a graph $G = (V, E)$ is a set $D \subseteq V$ such that each vertex $v \in V \setminus D$ of at least one neighbor in the set $D$.

The equivalence of conditions (i) and (ii) follows directly from the definition of codominant set. Conditions (ii) and (iii) are equivalent because the condition $D \not\subseteq N_G(v)$ is equivalent to the condition that the difference of sets $D \setminus N_G(v)$ is not empty. The equivalence of conditions (iii) and (iv) follows directly from the definition of the complement of a graph. The equivalence of conditions (iv) and (v) follows directly from the definition of the dominant set.

So we have shown that for an arbitrary set $D \subseteq V$, $D$ is a codominatng set in the graph $G$ if and only if it is a dominating set in the graph $\overline{G}$. In the lectures, we showed that the problem Dominating Set is NP-complete. The proof of NP-completeness of the problem Codominating Set now follows:

(a) The problem Codominating Set is in NP. Namely, in case of a yes instance, we can take any codominating set $D$ of size at most $k$ as a certificate; whether $D$ is indeed a codominating set and $|D| \leq k$ can be checked in polynomial time.

(b) The problem is NP-complete, since Dominating Set $\propto$ Codominating set:

For a given instance $I = (G = (V, E), k)$ of Dominating Set we can construct in polynomial time an instance $J = (\overline{G}, k)$ of Codominating Set.

Let us justify the correctness of the reduction. Since the graph $G$ is the complement of the graph $\overline{G}$, for any set $D \subseteq V$ it is true that $D$ is a dominating set in the graph $G = \overline{\overline{G}}$ if and only if $D$ is a codominating set in the graph $\overline{G}$. It follows that in the graph $G$ there exists a dominating set $D \subseteq V$ such that $|D| \leq k$ if and only if in the graph $\overline{G}$ there exists a codominating set $D \subseteq V$ such that $|D| \leq k$. With this, we justified that the mapping $I \mapsto J$ described above is indeed a polynomial reduction of the Dominating Set problem to the Codominating Set problem.

Since the Dominating Set problem is NP-complete, it follows that the Codominating Set problem is NP-complete as well.

5. *(Proving NP-completeness; suggested.)*

Show that the following problem is NP-complete.

| Connected Dominating Set | |
|---|---|
| **Input**: | A connected graph $G = (V, E)$, an integer $k$. |
| **Question:** | Is there a dominating set $D \subseteq V$ in the graph $G$ such that $|D| \leq k$ and the subgraph of the graph $G$ induced by the set $D$ is connected? |

*Hint: Recall the proof of NP-completeness of the Dominating Set problem given in class.*

**Solution:** Let us say that a dominating set $D$ in a graph $G$ is *connected* if the subgraph of the graph $G$ induced by the set $D$ is connected. The Connected Dominating Set problem is in NP: for a given connected dominating set $D \subseteq V$ of cardinality at most $k$ we can verify all these properties in polynomial time.

To prove NP-completeness, we follow the hint. In the lectures, we proved the NP-completeness of the Dominating Set problem with the following reduction from Vertex Cover:

using an instance $I = (G = (V, E), k)$ of the Vertex Cover problem, we computed an instance $J(I) = (G' = (V', E'), k)$ for the Dominating Set problem where:

- $V' = V \cup E$,

- $V$ is a clique in $G'$,
- $E$ is an independent set in $G'$,
- $ve \in E(G')$ if and only if $v \in V, e \in E, v$ is an endpoint of $e$.

Then the graph $G$ has a vertex cover of cardinality at most $k$ if and only if the graph $G'$ has a dominating set of cardinality at most $k$. In the proof of this equivalence, we constructed in both cases a dominating set $D$ of the graph $G'$ of cardinality at most $k$ such that $D \subseteq V$. Since $V$ is a clique in the graph $G'$, any such set $D$ is a connected dominating set of the graph $G'$.

It follows that a graph $G$ has a vertex cover of cardinality at most $k$ if and only if $G'$ has a connected dominating set of cardinality at most $k$. Using the above reduction from the VERTEX COVER problem, we therefore derive that the CONNECTED DOMINATING SET problem is NP-complete.

6. *(Proving* NP*-completeness; more challenging.)*

   Show that the following problem is NP-complete.

   | CLIQUE AND INDEPENDENT SET | |
   |---|---|
   | **Input:** | A graph $G = (V, E)$, an integer $k$. |
   | **Question:** | Does $G$ contain a clique of cardinality at least $k$ and an independent set of cardinality at least $k$? |

   *Hint: Make a reduction from the* CLIQUE *problem.*

   **Solution:** The CLIQUE AND INDEPENDENT SET problem is obviously in NP. In case of a yes instance, a certificate is given by a pair $(C, I)$ of sets of vertices of $G$ such that $C$ is a clique of cardinality at least $k$ and $I$ is an independent set of cardinality at least $k$. We can verify these properties in polynomial time.

   To prove NP-completeness, we follow the hint. Let $I = (G = (V, E), k)$ be an instance of the CLIQUE problem: Does $G$ contain a clique of cardinality at least $k$? From the graph $G$ let us construct in polynomial time a graph $G'$, for which it will hold that $G$ contains a clique of cardinality at least $k$ if and only if $G'$ contains a clique of cardinality at least $k$ and an independent set of cardinality at least $k$. If we could achieve that $G'$ will in every case contain an independent set of cardinality at least $k$, the above equivalence would simplify to the following equivalence: $G$ contains a clique of cardinality at least $k$ if and only if $G'$ contains a clique of cardinality at least $k$.

   The desired condition can be achieved by taking for $G'$ the graph obtained from the graph of $G$ by adding $k$ isolated vertices. For this process to be polynomial, we must ensure that the number $k$ – otherwise represented by $\log k$ bits – is at most polynomial in size depending on the number of vertices of the graph $G$. But of course we can assume that $k \leq |V(G)|$, since otherwise the CLIQUE problem can be solved trivially, with a negative answer.

   Regardless of the structure of the graph $G$, therefore, the graph $G'$ always has an independent set of cardinality at least $k$ (consisting of the set of $k$ isolated vertices in $V(G') \backslash V(G)$). Since isolated vertices are not contained in any clique of cardinality more than one, we observe that a graph $G$ contains a clique of cardinality at least $k$ if and only if $G'$ contains a clique of cardinality at least $k$. We conclude that $G$ contains a clique of cardinality at least $k$ if and only if $G'$ contains a clique of cardinality at least $k$ and an independent set of cardinality at least $k$. So we have described a valid polynomial reduction. Since the Clique problem NP-complete, the NP-completeness of the Clique and Independent Set problem follows.